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CSC_360
Class Notes

13/8/26

Topics Learned:

- Setup for Java programming and Maven.
- Setup of the environment required for Graphics Programming in Java.
- Setup of GitHub Repository using Git and Command Prompt.
- Creation and running a basic Java Program which creates Geometric square.

Java Setup and Validation:

- Installation and validation of the Java Development Kit (JDK).
- Validation of java and Java Compiler was performed by commands:
 - ~ java -version
 - ~ javac -version
 - ~ javac – Java compiler.
 - ~ java – Used to run compiled Java programs.

Maven Setup:

- Installation and Setup of Apache Maven.
- Apache Maven is used for building and managing Java Projects.
- Installation and validation of Apache Maven was done using:
 - ~ mvn -version
- Apache Maven Setup was associated with the Java JDK.

Setting Up the GitHub Repository:

- Used an already existing GitHub repository.

- Cloning of the repository on the local machine was done through the following command:
~ git clone <repository-url>
- Accessing the cloned repository was done via the local machine via the Command Prompt interface.
- The file created in the class exercise was saved in the cloned repository.

Java Graphics Application:

- A simple Java program that draws a square through lines was created.
- Java Swing and AWT Graphics were used to create the graphics window and the square.
- JFrame was used to create the graphics window for the application.
- JPanel was used as the drawing canvas.
- Graphics2D was used in drawing the graphic objects.
- drawRect() method was used in drawing the square.
- BasicStroke was used in creating a solid and unbroken line for the square.

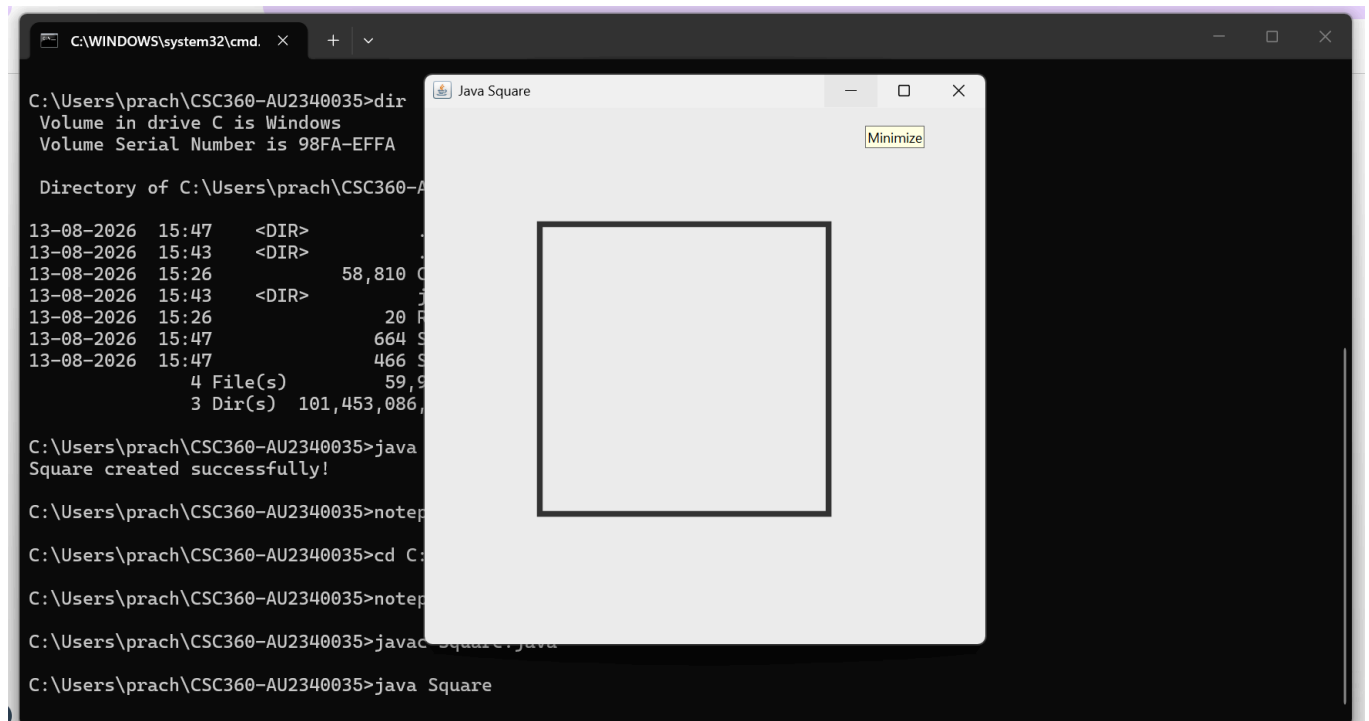
Compilation and Execution of the Program:

- Java source code was written as: Square.java
- The compilation of the program was done using: javac Square.java
- Execution of the program was done using: java Square
- Running the program opened a window displaying the square in graphical form.

Git Commands Used:

- After writing the Java program and ensuring that it runs, the files were then committed into the Git repository.
- Checking status of repository: git status

- Adding files: `git add .`
- Setting up Git identity if required: `git config --global user.name ,git config --global user.email`
- Committing changes: `git commit -m "Add Java square program"`
- Pushing to GitHub: `git push origin main`



The screenshot shows a Windows command prompt window with the title 'C:\WINDOWS\system32\cmd.' and a 'Java Square' application window. The command prompt displays the following commands and output:

```
C:\Users\prach\CSC360-AU2340035>dir
Volume in drive C is Windows
Volume Serial Number is 98FA-EFFA

Directory of C:\Users\prach\CSC360-AU2340035

13-08-2026  15:47    <DIR>
13-08-2026  15:43    <DIR>
13-08-2026  15:26         58,810  C
13-08-2026  15:43    <DIR>
13-08-2026  15:26         20 F
13-08-2026  15:47         664 S
13-08-2026  15:47         466 S
                4 File(s)      59,9
                3 Dir(s)  101,453,086,

C:\Users\prach\CSC360-AU2340035>java
Square created successfully!

C:\Users\prach\CSC360-AU2340035>notepad

C:\Users\prach\CSC360-AU2340035>cd C:

C:\Users\prach\CSC360-AU2340035>notepad

C:\Users\prach\CSC360-AU2340035>javac Square.java

C:\Users\prach\CSC360-AU2340035>java Square
```

The 'Java Square' application window is a simple gray window with a title bar and a 'Minimize' button. It contains a large, empty square frame.



PrachiSanaliya Add Java square program

Code

Blame

37 lines (26 loc) · 862 Bytes

```
1  import javax.swing.*;
2  import java.awt.*;
3
4  ✓ public class Square extends JPanel {
5
6      @Override
7  ✓  protected void paintComponent(Graphics g) {
8      super.paintComponent(g);
9
10     Graphics2D g2 = (Graphics2D) g;
11
12     // Make the lines smooth
13     g2.setRenderingHint(
14         RenderingHints.KEY_ANTIALIASING,
15         RenderingHints.VALUE_ANTIALIAS_ON
16     );
17
18     // Solid line with thickness 5 pixels
19     g2.setStroke(new BasicStroke(5));
20
21     // Draw a complete square
22     g2.drawRect(100, 100, 250, 250);
23 }
24
25 ✓ public static void main(String[] args) {
26
27     JFrame frame = new JFrame("Java Square");
28
29     frame.add(new Square());
30
31     frame.setSize(500, 500);
32     frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
33     frame.setLocationRelativeTo(null);
34
35     frame.setVisible(true);
36 }
37 }
```